Supplementary Material

ThermoSim: pinch-constrained solving of thermodynamic cycles in Python

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S1. Scope

This document accompanies the manuscript *ThermoSim: pinch-constrained solving of thermodynamic cycles in Python*, submitted to *SoftwareX*. It gives the complete enumeration of solvable and refused heat exchanger specifications referred to in Section 2.2 of the manuscript, where only the summary of refusal causes is reproduced.

Table S1 lists all fifteen two-unknown combinations of the six quantities that describe a counter-flow exchanger at fixed pressures, together with the free variable used by the root-finder and the relation that closes each case. Table S2 lists all twenty three-unknown combinations, giving for each the anchor available at either terminal, the unknown that anchor resolves, and the two-unknown case of Table S1 to which the specification then reduces. Table S3 groups the refusals by cause.

The enumeration is implemented in ThermoSim v3.3.0 (https://github.com/Nouman090/ThermoSim, DOI: 10.5281/zenodo.22156709) and is exercised by the package test suite; the same tables are maintained at the project wiki (https://github.com/Nouman090/ThermoSim/wiki).

Section S8 gives the full specification of the combined-cycle designs used in Sections 1 and 3 of the manuscript.

S2. Notation

Both operating pressures are taken as specified, so the temperature of either stream follows from its enthalpy and pressure. Six quantities then describe the exchanger: four terminal specific enthalpies and two mass flow rates.

Symbol	Meaning
H_{hi}, H_{ho}	specific enthalpy of the hot stream at inlet and at outlet
H_{ci}, H_{co}	specific enthalpy of the cold stream at inlet and at outlet
$\dot{m}_h$, $\dot{m}_c$	mass flow rate of the hot and of the cold stream
ΔH_h, ΔH_c	$H_{hi} - H_{ho}$ and $H_{co} - H_{ci}$
Q_h, Q_c	hot-side and cold-side duty, $\dot{m}_h \Delta H_h$ and $\dot{m}_c \Delta H_c$
ΔT_{pp}	specified minimum temperature approach (pinch)
T_x	temperature at the state whose specific enthalpy is H_x

S3. The three closure equations

Energy balance. $\dot{m}_h \Delta H_h = \dot{m}_c \Delta H_c$. One equation, and therefore one unknown, for any exchanger with both sides modelled.

Pinch equation. $\min(T_h - T_c)$ evaluated over the whole profile $= \Delta T_{pp}$. Imposing the specified approach as an equality rather than an inequality supplies a second equation, closing a second unknown. The profile is rebuilt at each trial from a discretisation into user-selectable divisions, into which the exact phase-boundary enthalpies are inserted so that no kink in the profile can fall between divisions.

Anchor relation. Where the pinch is known to occur at a particular terminal, a known temperature there fixes the temperature facing it across the approach, closing a third unknown. Anchors are always terminal, since only a corner carries a known temperature; an internal pinch at a bubble or dew point is located by the pinch equation rather than anchored. A specification with four or more unknowns is under-determined against these three equations and is refused without further analysis.

S4. Anchor relations and their admissibility

At the hot end the hot inlet faces the cold outlet; at the cold end the hot outlet faces the cold inlet. The four anchor relations are those of Figure 2(a) of the manuscript:

Terminal	Relations	Resolves
Hot end	$T_{co} = T_{hi} - \Delta T_{pp}$ $T_{hi} = T_{co} + \Delta T_{pp}$	whichever of H_{hi}, H_{co} is unknown
Cold end	$T_{ci} = T_{ho} - \Delta T_{pp}$ $T_{ho} = T_{ci} + \Delta T_{pp}$	whichever of H_{ho}, H_{ci} is unknown

An anchor at a terminal is available only when exactly one of the two enthalpies facing each other there is unknown: if both are known the anchor is redundant, and if both are unknown it cannot be applied. Which terminal may be anchored also depends on the exchanger type, because the location at which the closest approach can occur does:

Exchanger type	Possible pinch locations	Admissible anchor
Counter-flow (double pipe)	either terminal	hot end or cold end
Evaporator	hot-inlet end, or internally at the bubble point	hot end
Condenser	hot-outlet end, or internally at the dew point	cold end

Anchors are checked for physical consistency before they are applied, so that one which would place the cold inlet above the cold outlet is rejected rather than solved. A three-unknown specification solved this way attains the specified approach at two locations, the anchored terminal and the one located by the pinch equation, and therefore returns a double-pinch design.

S5. Enumeration of the two-unknown combinations

***Table S1**: Solvability of the fifteen two-unknown combinations. The free variable is the quantity on which the root-finder operates; refused cases have none.*

Sl	Unknowns	Free variable	Closed by / reason refused	Solvable
1	H_{hi}, H_{ho}	—	hot-side ΔH unknown, side duty undefined	✗
2	H_{ci}, H_{co}	—	cold-side ΔH unknown, side duty undefined	✗
3	H_{hi}, $\dot{m}_h$	$\dot{m}_h$	$H_{hi} = H_{ho} + Q_c/\dot{m}_h$	✓
4	H_{ho}, $\dot{m}_h$	$\dot{m}_h$	$H_{ho} = H_{hi} - Q_c/\dot{m}_h$	✓
5	H_{ci}, $\dot{m}_c$	$\dot{m}_c$	$H_{ci} = H_{co} - Q_h/\dot{m}_c$	✓
6	H_{co}, $\dot{m}_c$	$\dot{m}_c$	$H_{co} = H_{ci} + Q_h/\dot{m}_c$	✓

Sl	Unknowns	Free variable	Closed by / reason refused	Solvable
7	H_{hi}, H_{ci}	T_{hi}	$H_{ci} = H_{co} - \dot{m}_h \Delta H_h / \dot{m}_c$	✓
8	H_{hi}, H_{co}	—	knowns confined to the cold end, far-end profile unconstrained	✗
9	H_{ho}, H_{ci}	—	knowns confined to the hot end, far-end profile unconstrained	✗
10	H_{ho}, H_{co}	T_{ho}	$H_{co} = H_{ci} + \dot{m}_h \Delta H_h / \dot{m}_c$	✓
11	H_{hi}, $\dot{m}_c$	T_{hi}	$\dot{m}_c = \dot{m}_h \Delta H_h / \Delta H_c$	✓
12	H_{ho}, $\dot{m}_c$	T_{ho}	$\dot{m}_c = \dot{m}_h \Delta H_h / \Delta H_c$	✓
13	H_{ci}, $\dot{m}_h$	T_{ci}	$\dot{m}_h = \dot{m}_c \Delta H_c / \Delta H_h$	✓
14	H_{co}, $\dot{m}_h$	T_{co}	$\dot{m}_h = \dot{m}_c \Delta H_c / \Delta H_h$	✓
15	$\dot{m}_h$, $\dot{m}_c$	—	only the ratio $\dot{m}_h / \dot{m}_c$ is fixed	✗

Ten of the fifteen combinations are solvable.

S6. Enumeration of the three-unknown combinations

***Table S2**: Solvability of the twenty three-unknown combinations. For each terminal the table gives the unknown that the anchor there resolves and the case of Table S1 to which the specification then reduces. A dash means that no anchor is available at that terminal, either because both facing enthalpies are known or because both are unknown.*

Sl	Unknowns	Hot-end anchor	Reduces to	Cold-end anchor	Reduces to	Solvable
1	H_{hi}, H_{ho}, H_{ci}	H_{hi}	case 9	—	—	✗
2	H_{hi}, H_{ho}, H_{co}	—	—	H_{ho}	case 8	✗
3	H_{hi}, H_{ho}, $\dot{m}_h$	H_{hi}	case 4	H_{ho}	case 3	✓
4	H_{hi}, H_{ho}, $\dot{m}_c$	H_{hi}	case 12	H_{ho}	case 11	✓
5	H_{hi}, H_{ci}, H_{co}	—	—	H_{ci}	case 8	✗
6	H_{ho}, H_{ci}, H_{co}	H_{co}	case 9	—	—	✗
7	H_{ci}, H_{co}, $\dot{m}_h$	H_{co}	case 13	H_{ci}	case 14	✓
8	H_{ci}, H_{co}, $\dot{m}_c$	H_{co}	case 5	H_{ci}	case 6	✓
9	H_{hi}, H_{ci}, $\dot{m}_h$	H_{hi}	case 13	H_{ci}	case 3	✓
10	H_{hi}, H_{ci}, $\dot{m}_c$	H_{hi}	case 5	H_{ci}	case 11	✓
11	H_{ho}, H_{ci}, $\dot{m}_h$	—	—	—	—	✗
12	H_{ho}, H_{ci}, $\dot{m}_c$	—	—	—	—	✗
13	H_{hi}, H_{co}, $\dot{m}_h$	—	—	—	—	✗
14	H_{hi}, H_{co}, $\dot{m}_c$	—	—	—	—	✗
15	H_{ho}, H_{co}, $\dot{m}_h$	H_{co}	case 4	H_{ho}	case 14	✓
16	H_{ho}, H_{co}, $\dot{m}_c$	H_{co}	case 12	H_{ho}	case 6	✓
17	H_{hi}, $\dot{m}_h$, $\dot{m}_c$	H_{hi}	case 15	—	—	✗
18	H_{ho}, $\dot{m}_h$, $\dot{m}_c$	—	—	H_{ho}	case 15	✗
19	H_{ci}, $\dot{m}_h$, $\dot{m}_c$	—	—	H_{ci}	case 15	✗
20	H_{co}, $\dot{m}_h$, $\dot{m}_c$	H_{co}	case 15	—	—	✗

Eight of the twenty combinations are solvable. Every one of those eight admits an anchor at *both* terminals, so the count is eight for every exchanger type; what the type determines is which anchor is used, and hence which two-unknown case the specification reduces to. An evaporator takes the hot-end column, a condenser the cold-end column, and a counter-flow exchanger either. Case 16 solved as an evaporator, for example,

anchors H_{co} at the hot-inlet end and reduces to case 12, the same case as the organic Rankine cycle evaporator of Section 3 of the manuscript.

S7. Refusals grouped by cause

***Table S3**: The refused specifications grouped by cause. Rather than returning an untrusted root, the solver reports which of these four conditions applies.*

Cause of refusal	Table S1	Table S2	Total
Side duty undefined — both enthalpies on one side are unknown, so that side's ΔH, and therefore the duty, cannot be formed	1, 2	—	2
Far-end profile unconstrained — the known enthalpies are confined to one terminal, leaving the profile at the other end unbounded	8, 9	1, 2, 5, 6	6
Mass flow ratio only — the energy balance fixes $\dot{m}_h/\dot{m}_c$ but neither rate individually	15	17, 18, 19, 20	5
No admissible anchor — one terminal pair is fully known and the other fully unknown, so no terminal relation can be applied	—	11, 12, 13, 14	4

Five of fifteen two-unknown and twelve of twenty-three-unknown specifications are refused. In each case the solver reports the cause rather than returning a value.

S8. The combined-cycle model of Sections 1 and 3

Section 1 of the manuscript motivates the profile check with two heat-recovery steam generator designs that differ only in steam pressure and stack temperature. This section gives the full specification so that both may be reproduced.

Both plants share an identical gas-turbine topping cycle. Air enters the compressor at 1 bar and 300 K at 100 kg/s, is compressed through a pressure ratio of 12 at an isentropic efficiency of 0.86, heated to 1400 K, and expanded to 1 bar at an isentropic efficiency of 0.89, leaving the gas turbine at 552.74 °C. Combustion is represented as a specified heat input of 85.47 MW between the compressor discharge and the turbine inlet, consistent with the limitation stated in Section 4.4 of the manuscript. Because the topping cycle is common to both designs, the gas turbine work of 66.58 MW, the compressor work of 36.17 MW and the heat input are identical in the two cases, and the efficiency difference between them arises entirely in the bottoming cycle.

The bottoming cycle is a steam Rankine cycle condensing at 0.008 MPa, with a feed pump at an isentropic efficiency of 0.80 and a steam turbine at 0.88. Steam leaves the heat-recovery steam generator at 500 °C in both designs. The generator is modelled as a counter-flow exchanger with a specified minimum approach of 15 K, discretised into 200 divisions for the reported profiles. Its hot inlet is the gas turbine exhaust and its hot outlet the stack, whose temperature is specified; its cold inlet is the pump discharge and its cold outlet the steam turbine inlet. The steam mass flow rate is the single remaining unknown and is returned by the energy balance.

The two designs differ only in the pair (steam pressure, stack temperature). Table S4 gives the resulting operating points.

***Table S4**: The two combined-cycle designs of Section 1. The topping cycle is common to both; only the live steam pressure and the stack temperature differ. Both terminal temperature differences are positive in each design, and in Plant A both exceed the specified 15 K approach.*

Quantity	Plant A	Plant B
Live steam pressure (bar)	60	40
Stack temperature (°C)	120	150
Steam mass flow rate (kg/s)	14.086	13.044
Heat-recovery steam generator duty (MW)	45.662	42.617
Feedwater inlet temperature (°C)	42.05	41.87
Steam turbine work (MW)	15.743	14.084
Feed pump work (kW)	106.3	65.6
Net plant power (MW)	46.038	44.421
Thermal efficiency (%)	53.87	51.98
Terminal difference, hot end (K)	+52.74	+52.74
Terminal difference, cold end (K)	+77.95	+108.13
Minimum approach over the profile (K)	**−13.55**	**+15.09**

The violation in Plant A is interior to the exchanger and occurs at the water-side bubble point, where the gas has fallen to 262.03 °C while the water has reached its saturation temperature of 275.59 °C at 60 bar. Raising the steam pressure raises that saturation temperature, and lowering the stack temperature raises the steam flow rate and so steepens the gas profile; the two changes together drive the gas below the evaporating water without either terminal giving any indication.

The reported approach does not depend on the discretisation. Inserting the exact phase-boundary enthalpies into the division grid returns −13.5512 K for Plant A and +15.0876 K for Plant B at every division count from 50 to 1000, in both cases equal to the value obtained analytically at the bubble point.

The script that builds both plants and generates Figure 1 of the manuscript is distributed with the code release (C2).